

Gravitational waves scattering off a spherical thin shell

Summary of the project: junction conditions, reflectivity, quasinormal modes and stability

Final project, gravitational-wave course

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Abstract

A thin spherical shell of matter at radius $R > 2M$ separates a flat interior from a Schwarzschild exterior of mass M . It is the simplest horizonless compact object. We compute, for odd (axial) and even (polar) gravitational perturbations, the reflection and transmission coefficients $\mathcal{R}(\omega)$ and $\mathcal{T}(\omega)$ and the quasinormal-mode (QNM) spectrum. The first step is to find a background that can be static at all. A domain wall cannot be; a fluid shell with a specific positive pressure can. For odd parity the shell then turns out to be a repulsive δ -function in the wave equation. For even parity it is a flux-conserving 2×2 transfer matrix. The shell is opaque at low frequency and transparent at high frequency, and it approaches the black-hole reflectivity as $R \rightarrow 2M$. Its ringing is completely different from a black hole's, with long-lived trapped modes for $R < 3M$. One even-parity mode of the shell's own matter grows exponentially, so the static shell is unstable, in agreement with Pitre, Schneider and Poisson (2026). Units: $G = c = 1$, time dependence $e^{-i\omega t}$.

1 The problem and the first obstacle: is there a static shell?

Inside the shell, Birkhoff's theorem and regularity at $r = 0$ force flat space: $ds^2 = -dT^2 + dr^2 + r^2 d\Omega^2$. Outside, $ds^2 = -f dt^2 + f^{-1} dr^2 + r^2 d\Omega^2$ with $f = 1 - 2M/r$. The shell carries a surface stress tensor $S_{ab} = \sigma u_a u_b + p(h_{ab} + u_a u_b)$. Gluing the two geometries is governed by the Israel junction conditions [4, 6]:

$$[h_{ab}] = 0, \quad [K_{ab}] - h_{ab}[K] = -8\pi S_{ab}, \quad [X] \equiv X_+ - X_-, \quad (1)$$

where h_{ab} is the induced metric and K_{ab} the extrinsic curvature. For a spherical shell $r = R(\tau)$ they give the mass formula and the equation of motion (Ipser & Sikivie [5], Eqs. 3.7-3.9):

$$\sigma = \frac{\sqrt{1 + \dot{R}^2} - \sqrt{f(R) + \dot{R}^2}}{4\pi R}, \quad (\alpha + \beta)\ddot{R} = -\frac{\alpha M}{R^2} + \frac{2p}{\sigma} \frac{\alpha\beta(\alpha + \beta)}{R}, \quad (2)$$

with $\alpha = \sqrt{1 + \dot{R}^2}$ and $\beta = \sqrt{f + \dot{R}^2}$.

Idea. A monochromatic scattering problem ($e^{-i\omega t}$) needs a time-independent background. Before any wave calculation we must therefore ask whether the shell can sit still.

Gravity always pulls inward ($-\alpha M/R^2$). Pressure pushes outward only if $p > 0$. Setting $\dot{R} = \ddot{R} = 0$ gives a unique answer:

$$\frac{p}{\sigma} = \kappa(R) = \frac{1 - \sqrt{f_R}}{4\sqrt{f_R}} > 0, \quad \sigma = \frac{1 - \sqrt{f_R}}{4\pi R}. \quad (3)$$

This leads to four candidate models:

- **Domain wall** ($p = -\sigma$, the object of the prior notes). $\ddot{R} < 0$ always, so it collapses to a black hole [5]. It can only be *frozen* at its turning point (*model A*), where $\dot{R} = -(1 + 3\sqrt{f_R})/2R$.
- **Dust** ($p = 0$). It is not static either: $(\alpha + \beta)\ddot{R} = -\alpha M/R^2 < 0$.
- **Static perfect fluid** with $p = \kappa(R)\sigma$ (*model C*, the primary model). The pressure ratio is fixed by the compactness, not free. The energy condition $p \leq \sigma$ requires $R \geq 25M/12$. For the perturbed shell we need an equation of state, $\delta p = \Gamma \frac{p}{\sigma + p} \delta \sigma$ with $\Gamma = 2$, as in [15].
- **Gravastar** (de Sitter interior [12]). Not treated.

The interior clock runs slower: $dT = \sqrt{f_R} dt$. A wave of frequency ω outside therefore has frequency $\nu = \omega/\sqrt{f_R}$ inside.

2 Waves on each side and the junction conditions

Idea. Away from the shell everything is textbook. Outside, the Regge–Wheeler/Zerilli equations hold; inside, the flat wave equation. *All* the physics of the shell is in how the two solutions are glued at $r = R$.

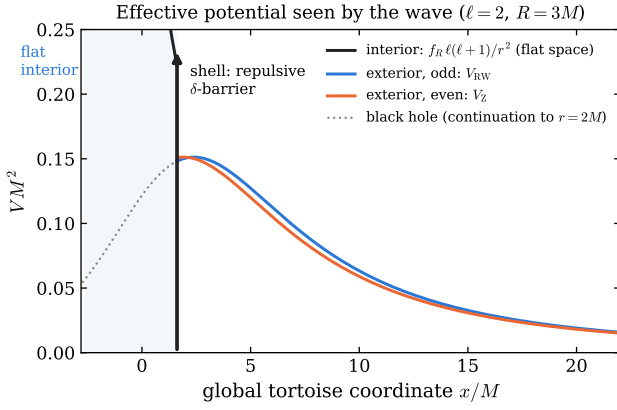


Figure 1: Potential seen by the wave. Flat interior (centrifugal barrier), shell (δ -barrier, arrow) and exterior RW/Zerilli barrier.

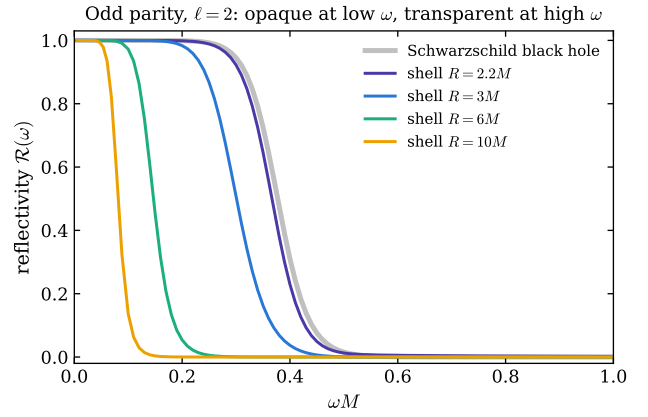


Figure 2: Reflectivity for odd parity, $\ell = 2$. Compact shells approach the black-hole curve.

We use the gauge-invariant master functions of Martel and Poisson [7]: Cunningham–Price–Moncrief for odd parity and Zerilli–Moncrief for even parity. Outside, $\Psi_{,r_*r_*} + [\omega^2 - V]\Psi = 0$, with $V_{\text{RW}} = f[\ell(\ell+1)/r^2 - 6M/r^3]$ [1] or the Zerilli potential V_Z [2]. Inside, $\Psi'' + [\nu^2 - \ell(\ell+1)/r^2]\Psi = 0$. Its regular solution is the Riccati–Bessel function $\hat{j}_\ell(\nu r)$. In a global tortoise coordinate x ($x = r_*$ outside, $dx = dr/\sqrt{f_R}$ inside) the whole problem is a 1D wave equation with the potential of Fig. 1.

The linearized Israel equations (1) were derived from first principles. On each side the metric perturbation is written in its own Regge–Wheeler gauge. Because those gauges are not adapted to the shell, the shell’s position is described by perturbed embeddings, with radial displacements $\xi_\pm$ that differ on the two sides. The metric perturbations are then expressed through the master functions. The derivation was done symbolically (SymPy) and verified. The results are the following.

Odd parity: a δ -barrier. Odd perturbations do not move the shell or its matter. The background term $[K]^{(0)}$, flagged in the prior notes as the missing step, cancels exactly against the pressure term. What remains is

$$\Psi_+ = \Psi_-, \quad \left[\frac{d\Psi}{dx} \right] = \Delta \Psi, \quad \Delta = \frac{\sqrt{f_R}(1 - \sqrt{f_R})}{R} = 4\pi\sigma\sqrt{f_R} > 0. \quad (4)$$

The shell acts on axial waves as a repulsive $\delta(x - x_R)$ potential. Its strength does not depend on ω , on the equation of state, or on whether the shell is a wall or a fluid. The same condition was recovered independently by letting a smooth anisotropic shell become thin; the mismatch vanishes linearly in the thickness (Fig. 8). It also agrees with [15] and with the matching conditions of [12].

Even parity: a 2×2 transfer matrix. Polar perturbations move the shell radially and make its fluid oscillate. After eliminating the shell variables ($\xi_\pm$, $\delta\sigma$, velocity) one gets

$$\begin{pmatrix} \Psi_+ \\ \Psi'_+ \end{pmatrix}_R = \mathsf{T}(\omega; \ell, R/M, \Gamma) \begin{pmatrix} \Psi_- \\ \Psi'_- \end{pmatrix}_R, \quad \det \mathsf{T} = f_R^{-1/2} \text{ (exactly)}. \quad (5)$$

Ψ itself jumps across the shell, and T depends on frequency, with poles on the matter scale $\omega \sim (M/R^3)^{1/2}$. The determinant identity was not imposed; it is what guarantees energy conservation, and it holds to machine precision.

3 Scattering: how to define $\mathcal{R}$ and $\mathcal{T}$, and what they are

Idea. Nothing is absorbed, so the *closed* system reflects everything: $|B_{\text{out}}/B_{\text{in}}| = 1$. To ask how much the object reflects, we *open* the interior. A wave that crosses the shell is let go (only the ingoing Hankel wave $\hat{h}_\ell^{\text{in}}(\nu r)$ inside), exactly as a horizon lets it go for a black hole.

Outside, $\Psi \rightarrow B_{\text{in}}e^{-i\omega r_*} + B_{\text{out}}e^{i\omega r_*}$, and we define $\mathcal{R} = |B_{\text{out}}/B_{\text{in}}|^2$ and $\mathcal{T} = 1/|B_{\text{in}}|^2$. Flux conservation through the junction then implies $\mathcal{R} + \mathcal{T} = 1$. Figures 2–3 show four regimes:

- **Low frequency:** $\mathcal{R} \rightarrow 1$. Long waves are reflected, as for any compact object [14].
- **High frequency:** the potentials become negligible and only the δ -barrier is left. Then $\mathcal{R} \simeq \Delta^2/(4\omega^2 + \Delta^2) \rightarrow 0$: *the thin shell is transparent*. Even parity (model C) follows the same law, because the fluid’s inertia no longer matters (Fig. 3).
- **Compact shells, $R \rightarrow 2M$:** $\Delta \rightarrow 0$ and the shell sits deep in the redshifted throat, so $\mathcal{R} \rightarrow \mathcal{R}_{\text{BH}}$ (Vishveshwara [3]). The approach is slow, $\propto (R - 2M)^{1/2}$.

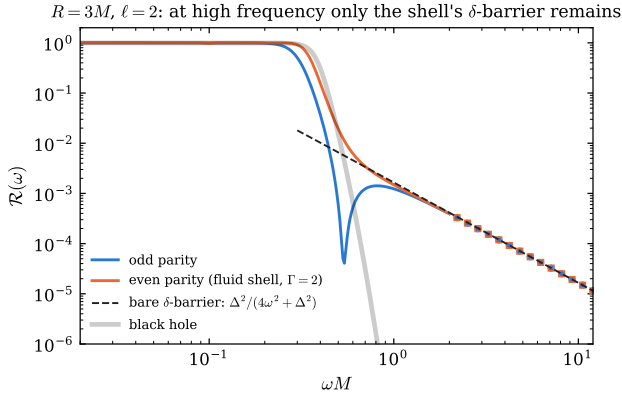


Figure 3: Odd and even ($\Gamma = 2$) reflectivity at $R = 3M$ on log axes, with the δ -barrier law. Dots: points computed up to $\omega M = 12$.

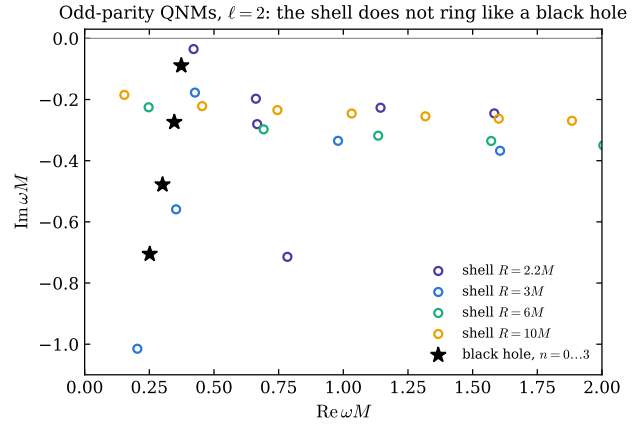


Figure 4: Odd QNMs ($\ell = 2$) for four compactnesses, against the Schwarzschild modes (stars).

- **Dilute shells:** the transition to transparency happens at $\omega R \sim \ell$, set by the flat-space centrifugal barrier. Even and odd reflectivities differ at intermediate frequencies. The shell breaks the isospectrality of Schwarzschild.

4 Quasinormal modes: how the shell rings

QNMs are the complex ω at which the regular interior solution, carried through the junction, is purely outgoing at infinity. They are found as zeros of a Wronskian. The outgoing solution comes from a Leaver-type continued fraction [8, 12] and independently from inward integration along a complex ray [14]. Three families appear (Figs. 4, 5):

1. **Wave modes** of the interior cavity, with $\omega \sim 1/R$. For dilute shells $\text{Re}(\omega R) \approx n\pi$ and $\text{Im}(\omega R) \approx -\frac{1}{2} \ln[2\ell(\ell+1)R/M]$ [15]. The cavity leaks through a weakly reflecting wall, so the damping is large and grows only logarithmically with R .
2. **Trapped modes** for $R < 3M$, between the flat interior and the exterior photon-sphere barrier. They are damped only by tunnelling and become extremely long-lived as $R \rightarrow 2M$: $\text{Im} \omega M \approx -10^{-6}$ at $R = 2.01M$. These are the modes behind gravitational-wave *echoes* [13]. *None of them tends to a Schwarzschild QNM*, even though the reflectivity does tend to the black-hole one.
3. **Matter modes** (even parity, model C only): oscillations of the shell itself, four per ℓ , with $\omega \sim (M/R^3)^{1/2}$. One of them is purely imaginary with $\text{Im} \omega > 0$.

5 The static shell is unstable

Idea. The only static member of the family, model C, should be stable if scattering off it is to describe something physical. The growing matter mode shows that it is not. It is then essential to compare time scales.

At $M/R = 0.2$, $\ell = 2$, $\Gamma = 2$ the growing mode is $\omega(R^3/M)^{1/2} = 0.6077i$, and the oscillating pair is $1.2671 - 0.0011i$. These values agree with Pitre, Schneider and Poisson [15] at every compactness (Fig. 5), although their method (retarded-time slicing, spectral solver) is completely different. As $M/R \rightarrow 0$ they tend to the oscillations of a *Newtonian* self-gravitating fluid shell. We derived that problem from scratch; it gives $\zeta^4 - S\zeta^2 + P = 0$ with $\zeta^2 = \omega^2 R^3/M$, and for $\ell = 2$ the roots are 1.5050 and $0.5147i$. The instability is therefore Newtonian in origin: the tangential pressure that holds the shell up makes $\ell \geq 2$ deformations run away. The same derivation shows that the constant term of the quartic in [15], Eq. (7.6), contains a misprint; their closed form, Eq. (7.8), is correct.

Consequence for the models (Fig. 6). The fluid shell grows on $\sim 1.6(R^3/M)^{1/2}$, which is much longer than the wave periods ($\sim R$). Scattering off model C is therefore a meaningful adiabatic idealization. The domain wall collapses on $\omega_{\text{dyn}}^{-1} \approx R/\sqrt{2}$, the light-crossing time, which is comparable to the wave periods. For the frozen wall (model A) the even-parity junction system is overdetermined and inconsistent (8 equations, 6 unknowns). Which two Israel equations are dropped is a free choice, and different choices change $\mathcal{R}$ at $O(1)$, from total reflection to the δ -law (Fig. 7). **The even-parity results of model A are not predictive;** its odd sector is identical to model C's.

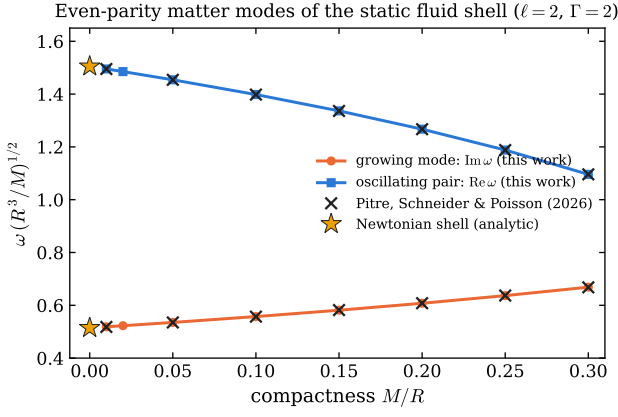


Figure 5: Matter modes of the fluid shell vs compactness: growing mode (orange) and oscillating pair (blue). Crosses: [15]. Stars: Newtonian shell.

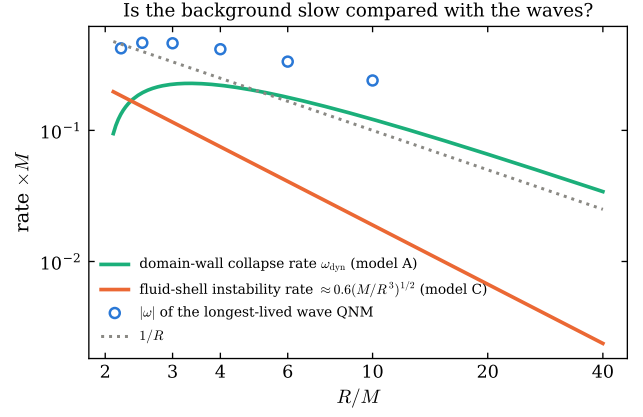


Figure 6: Background rates vs the least-damped wave QNM. The wall collapses as fast as the waves oscillate; the fluid instability is slow.

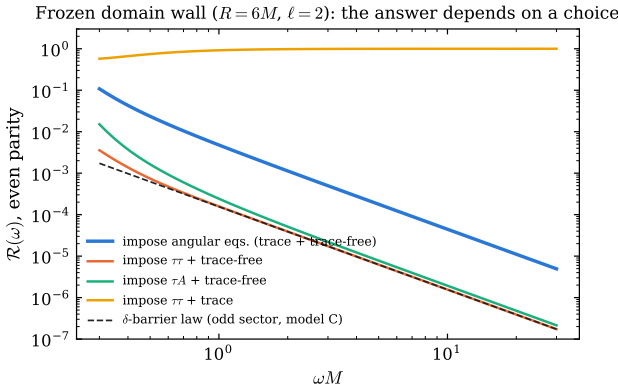


Figure 7: Frozen domain wall, even parity: four equally defensible choices of which Israel equations to impose give different $\mathcal{R}(\omega)$.

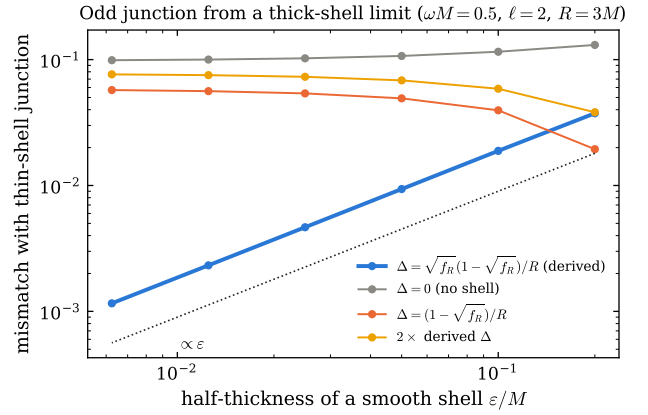


Figure 8: Independent test of the odd junction: a smooth shell of half-thickness ε converges to Eq. (4) and to no alternative.

6 How we know it is right

test	result
energy balance $ \mathcal{R} + \mathcal{T} - 1 $ (odd, even model C, $\omega M \leq 2$)	$\lesssim 6 \times 10^{-10}$
transfer matrix vs direct shooting, $\mathcal{R}$ and $\mathcal{T}$	$\lesssim 10^{-9}$
transfer matrix vs shooting, QNMs (397 modes)	median $\sim 10^{-11}$, worst 6×10^{-5}
Schwarzschild $\ell = 2$, $n = 0$ (Leaver)	$0.3736716844 - 0.0889623157i$ [11]
black-hole reflectivity at $(2M\omega)^2 = 1$	1.55×10^{-2} , as in [3]
WKB orders 1, 3, 6 vs exact ($\ell = 2$, $n = 0$)	relative errors 7%, 0.15%, 0.02% [9, 10]
odd junction from a thick-shell limit	converges $\propto$ thickness
$\det \mathbf{T} \sqrt{f_R} = 1$ (even, model C)	to 10^{-12}
time-domain ringdown vs QNM, $R = 2.2M$	$0.4215 - 0.0352i$ vs $0.4211 - 0.0350i$
time-domain ringdown vs QNM, black hole	$0.3731 - 0.0885i$ vs $0.3737 - 0.0890i$

7 Conclusions

1. A thin shell with a flat interior is static only if it has positive surface pressure, $p = \sigma(1 - \sqrt{f_R})/4\sqrt{f_R}$. Domain walls and dust shells collapse.
2. For axial waves the shell is a repulsive δ -potential, $\Delta = \sqrt{f_R}(1 - \sqrt{f_R})/R$. For polar waves it is a frequency-dependent, flux-conserving 2×2 transfer matrix, which also contains the dynamics of the shell's matter.
3. $\mathcal{R}(\omega)$ goes from total reflection at low frequency to transparency at high frequency ($\mathcal{R} \propto \omega^{-2}$). It approaches

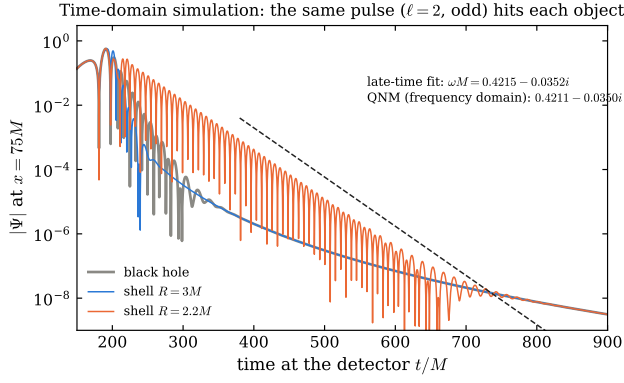


Figure 9: Time-domain simulation (odd, $\ell = 2$). The same pulse is sent onto a black hole and onto two shells, and the signal is recorded at $x = 75M$. The black hole rings down quickly. The $R = 2.2M$ shell keeps ringing at the frequency of its trapped QNM, found independently in the frequency domain. All curves end in the same late-time power-law tail of the exterior Schwarzschild potential.

the black-hole reflectivity as $R \rightarrow 2M$, while the QNM spectrum does not approach the black-hole one. Compact shells trap long-lived modes, which produce echoes.

4. The static shell is dynamically unstable through a Newtonian-origin matter mode. The instability is slow compared with the waves, so the scattering results for the fluid shell are physically meaningful. The frozen domain wall is not a controlled approximation in the even sector.

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